

SAT Diagnostic — Question Bank

40 Questions · 20 Math + 20 English · SAT Samarkand

■ MATH — 20 Questions

1. If $4x - 9 = 23$, what is the value of x ?

- (A) 6
- (B) 8
- (C) 11
- (D) 32

2. Which value of x satisfies $3(x + 4) = 2(x + 9)$?

- (A) 3
- (B) 6
- (C) -3
- (D) 12

3. A line passes through (2, 5) and (6, 13). What is its slope?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

4. If $f(x) = 2x^2 - 3x + 1$, what is $f(-2)$?

- (A) 11
- (B) 15
- (C) 3
- (D) -3

5. A car travels 240 km in 3 hours. At the same speed, how long will it take to travel 400 km?

- (A) 4 h
- (B) 4.5 h
- (C) 5 h
- (D) 5.5 h

6. What are the solutions to $x^2 - x - 12 = 0$?

- (A) $x = 3$ and $x = -4$
 - (B) $x = 4$ and $x = -3$
 - (C) $x = 6$ and $x = -2$
 - (D) $x = -6$ and $x = 2$
-

7. The sum of three consecutive even integers is 78. What is the largest?

- (A) 24
 - (B) 26
 - (C) 28
 - (D) 30
-

8. If 40% of a number is 56, what is 25% of that number?

- (A) 35
 - (B) 40
 - (C) 42
 - (D) 50
-

9. Simplify: $(x^2 - 9) / (x + 3)$, where $x \neq -3$.

- (A) $x - 3$
 - (B) $x + 3$
 - (C) $x^2 - 3$
 - (D) $x(x - 3)$
-

10. In a right triangle, the two legs measure 9 and 12. What is the length of the hypotenuse?

- (A) 13
 - (B) 14
 - (C) 15
 - (D) 17
-

11. The price of a jacket was reduced by 30% to \$91. What was the original price?

- (A) \$120
 - (B) \$125
 - (C) \$130
 - (D) \$140
-

12. If the system $2x + y = 11$ and $x - y = 1$ is solved, what is $x + y$?

- (A) 5
 - (B) 6
 - (C) 7
 - (D) 8
-

13. A bag has 4 red and 6 blue marbles. One is drawn at random. What is the probability it is red?

- (A) $\frac{2}{5}$
 - (B) $\frac{3}{5}$
 - (C) $\frac{1}{4}$
 - (D) $\frac{1}{3}$
-

14. What is the area of a circle with circumference 10π ?

- (A) 25π
 - (B) 50π
 - (C) 100π
 - (D) 5π
-

15. If $2^{(x+1)} = 32$, what is x ?

- (A) 3
 - (B) 4
 - (C) 5
 - (D) 6
-

16. A rectangle has area 60 cm^2 and width 5 cm. What is its perimeter?

- (A) 34 cm
 - (B) 38 cm
 - (C) 42 cm
 - (D) 46 cm
-

17. The mean of five numbers is 14. Four of them are 10, 12, 16, and 18. What is the fifth?

- (A) 12
 - (B) 13
 - (C) 14
 - (D) 15
-

18. Which inequality is equivalent to $-3x + 6 < 15$?

- (A) $x > -3$
 - (B) $x < -3$
 - (C) $x > 3$
 - (D) $x < 3$
-

19. A triangle has angles in the ratio 2 : 3 : 5. What is the measure of the largest angle?

- (A) 45°
 - (B) 72°
 - (C) 90°
 - (D) 108°
-

20. If p and q are positive integers and $p/q = 0.6$, which of the following could be $p + q$?

- (A) 8
 - (B) 9
 - (C) 10
 - (D) 11
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■ ENGLISH — 20 Questions

Passage 1

Passage 1 — Adapted from a science article (2023) Coral reefs cover less than one percent of the ocean floor yet support nearly one quarter of all marine species. Scientists describe them as the 'rainforests of the sea.' Rising ocean temperatures, however, are causing a process called bleaching, in which corals expel the algae living in their tissues. Without those algae, the corals turn white and, if stress continues, eventually die. Researchers are now experimenting with heat-resistant coral strains that may survive warmer waters.

Passage 2

Passage 2 — Adapted from a social-science essay (2024) For decades, economists assumed that people make decisions by carefully weighing costs and benefits. Behavioral economists have challenged this view. Their research shows that humans rely on mental shortcuts — called heuristics — that often lead to predictable errors. For instance, people tend to value a potential loss more strongly than an equivalent gain, a tendency known as loss aversion. Understanding these biases has practical applications: policymakers can design 'nudges' that steer people toward better choices without restricting their freedom.

1. Each of the students _____ required to submit a portfolio by Friday.

- (A) are
 - (B) is
 - (C) have been
 - (D) were
-

2. The committee has reached _____ decision after three days of debate.

- (A) their
 - (B) it's
 - (C) its
 - (D) they're
-

3. She studied for hours, _____ she still felt unprepared for the exam.

- (A) but
 - (B) and
 - (C) so
 - (D) for
-

4. The scientist presented her findings; _____, the audience asked many questions.

- (A) however
 - (B) afterward
 - (C) therefore
 - (D) otherwise
-

5. Neither the manager nor the employees _____ aware of the new policy.

- (A) was
 - (B) is
 - (C) has been
 - (D) were
-

6. As used in Passage 1, the word 'expel' most nearly means

- (A) absorb
 - (B) release
 - (C) destroy
 - (D) transform
-

7. In Passage 1, the phrase 'rainforests of the sea' is used primarily to

- (A) explain why coral reefs are endangered
 - (B) emphasize the biodiversity coral reefs support
 - (C) compare ocean and forest ecosystems
 - (D) argue for increased environmental funding
-

8. According to Passage 1, coral bleaching occurs when

- (A) algae grow too rapidly inside coral tissue
 - (B) rising temperatures cause corals to expel their algae
 - (C) corals are exposed to excessive sunlight
 - (D) scientists introduce heat-resistant strains
-

9. The author's mention of researchers experimenting with heat-resistant coral strains suggests

- (A) the problem of bleaching has already been solved
 - (B) scientists believe warmer oceans are inevitable
 - (C) there may be a technological solution to bleaching
 - (D) heat-resistant corals currently dominate the ocean
-

10. As used in Passage 2, 'heuristics' most nearly means

- (A) detailed calculations
 - (B) random decisions
 - (C) mental shortcuts
 - (D) economic models
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11. The main purpose of Passage 2 is to

- (A) prove that traditional economics is completely wrong
 - (B) describe how behavioral economics challenges classical economic theory
 - (C) argue that people should always follow expert advice
 - (D) explain why loss aversion is a dangerous tendency
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12. According to Passage 2, loss aversion means that people

- (A) refuse to take any financial risks
 - (B) respond more strongly to potential losses than to equivalent gains
 - (C) make decisions based purely on logic
 - (D) always choose the option with the highest reward
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13. Running at full speed, _____ crossed the finish line first.

- (A) the finish line was crossed by the athlete
 - (B) it was the athlete who
 - (C) the athlete
 - (D) having won
-

14. The researcher's conclusions, which were published last year, _____ widely debated.

- (A) is
 - (B) was
 - (C) are
 - (D) has been
-

15. Which revision BEST combines these three sentences into one? 'The new policy will help students. It will reduce stress. It will improve grades.'

- (A) The new policy will help students, it reduces stress, improving grades.
 - (B) The new policy will help students by reducing stress and improving grades.
 - (C) The new policy, reducing stress and improving grades, it will help students.
 - (D) Students will be helped because stress is reduced and grades are improved.
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16. Which transition BEST introduces a sentence that contrasts with the previous idea?

- (A) Furthermore,
 - (B) As a result,
 - (C) Nevertheless,
 - (D) For example,
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17. The CEO's _____ manner put investors at ease during the turbulent financial period.

- (A) erratic
 - (B) composed
 - (C) aggressive
 - (D) indifferent
-

18. The evidence in the report was _____, drawing from government records, scientific studies, and independent audits.

- (A) sparse
 - (B) anecdotal
 - (C) comprehensive
 - (D) subjective
-

19. Although the plan seemed _____ at first, closer examination revealed several serious flaws.

- (A) implausible
 - (B) viable
 - (C) irrelevant
 - (D) complex
-

20. The professor's lecture was so _____ that even students unfamiliar with the topic could follow it easily.

- (A) technical
 - (B) abstract
 - (C) accessible
 - (D) lengthy
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